# Empirical Proof Record: VALIDATED

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## Empirical Proof Record — VALIDATED

**Proof ID:** `d839a87f5772076323a335e6225637ad…`
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### 1. Claim

Across 30 synthetic (x, y) pairs of size 100 (seed=20260518), “scipy.stats.pearsonr“, “numpy.corrcoef“, and “pingouin.corr(method='pearson')“ produce Pearson r values that agree to ≤ 1e-09 across all three pairwise comparisons. (RFC 0002 Phase E1: cross-framework validation at the statistical-primitive layer; three-way variant.)

- **Operationalisation:** Generate N (x, y) pairs where x ~ Normal(0, 1) and y = $\rho$_target × x + sqrt(1 - $\rho$_target$^2$) × Normal(0, 1) for a sweep of $\rho$_target across [-0.9, 0.9]. Compute Pearson r under each of scipy, numpy, pingouin. Compute max_abs_diff = max over (pair, backend-comparison) of |r_a - r_b|. VALIDATED iff max_abs_diff ≤ tolerance across all three pairwise comparisons.
- **Threshold:** `max_absolute_pearson_difference <= 1e-09 correlation`
- **H0:** At least one backend pair (scipy↔numpy, scipy↔pingouin, numpy↔pingouin) disagrees on Pearson r by more than the floating-point tolerance.
- **H1:** All three backends compute identical Pearson r to floating-point tolerance across every pair.

### 2. Verdict

**VALIDATED** — observed `3.3306690738754696e-16`
30 pairs × 3 backends; max pairwise |Δr| = 3.331e-16 on scipy_vs_numpy (≤ 1e-09 tol)

## 3. Evidence

| Pillar | Statistic | Valu |
|---|---|---|
| pearson_scipy_vs_numpy_vs_pingouin | max_absolute_pearson_difference | 3.3306690738754696e-1 |

## 4. Pre-registration

- Registered at: 2026-05-20T15:34:47.265470+00:00
- Config hash: 395ced3ae94d681473ffeddcaa298fd2…
- Data hash: 395ced3ae94d681473ffeddcaa298fd2…

## 5. Signature

fe6533ba848321d8e36ef9243c9a871d6fb001528d9e3493b8a70d2612111250